



OPEN ACCESS

EDITED BY

Goutam Thakur,
Manipal Institute of Technology, India

REVIEWED BY

Lopamudra Das Ghosh,
Texas A&M University, United States
Aparna Datta,
NSHM Knowledge Campus, India

*CORRESPONDENCE

Kagya A. Amoako
✉ Kamoako@newhaven.edu

RECEIVED 21 August 2025

REVISED 22 November 2025

ACCEPTED 27 November 2025

PUBLISHED 04 February 2026

CITATION

Majin S, Malik A, Poudel P, Faustino EVS,
Zuverza-Mena N, Hwa J, Takyar S, Shea S and
Amoako KA (2026) Drug-loaded nanoparticles
reduced platelet aggregation and blood
coagulation.
Front. Med. Technol. 7:1690389.
doi: 10.3389/fmedt.2025.1690389

COPYRIGHT

© 2026 Majin, Malik, Poudel, Faustino,
Zuverza-Mena, Hwa, Takyar, Shea and
Amoako. This is an open-access article
distributed under the terms of the [Creative
Commons Attribution License \(CC BY\)](#). The
use, distribution or reproduction in other
forums is permitted, provided the original
author(s) and the copyright owner(s) are
credited and that the original publication in
this journal is cited, in accordance with
accepted academic practice. No use,
distribution or reproduction is permitted
which does not comply with these terms.

Drug-loaded nanoparticles reduced platelet aggregation and blood coagulation

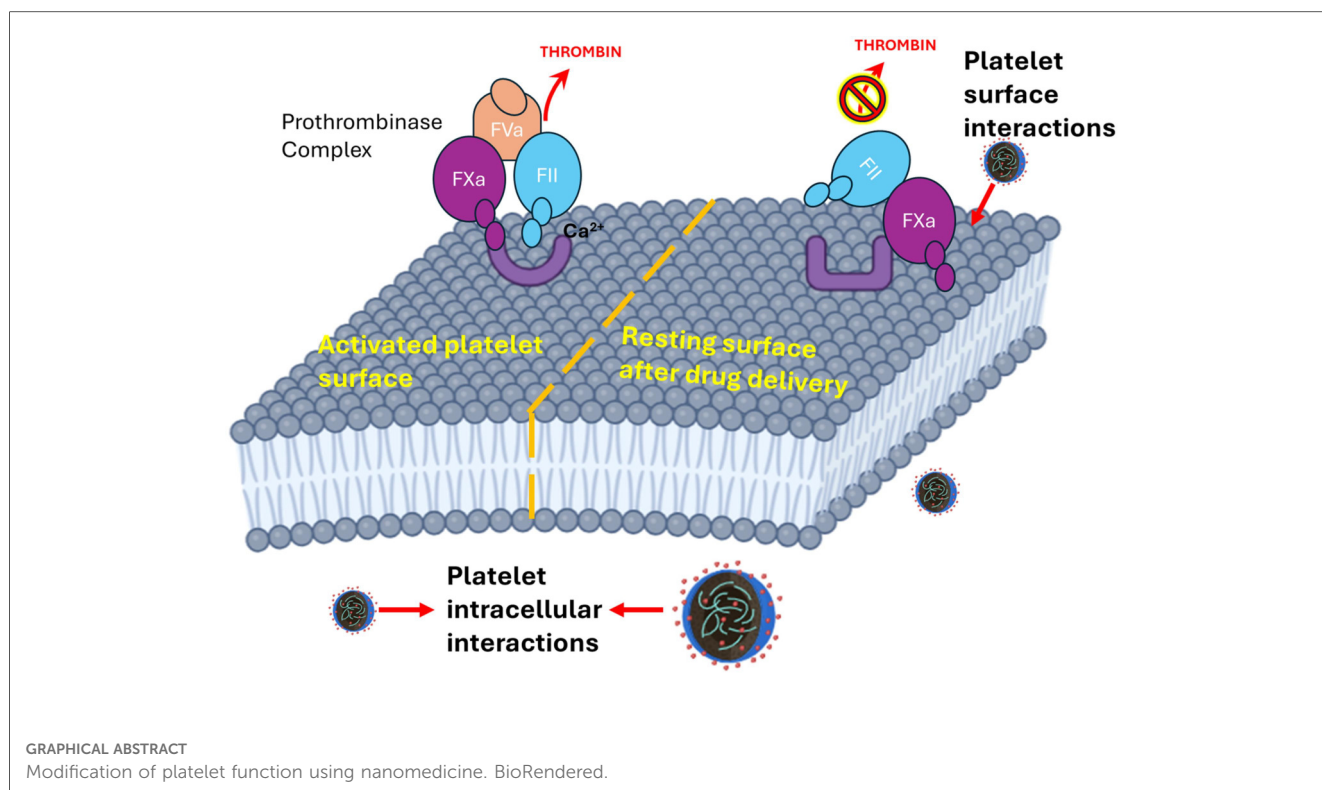
Sarah Majin¹, Afrida Malik¹, Pratima Poudel¹,
E. Vincent S. Faustino², Nubia Zuverza-Mena³, John Hwa⁴,
Seyedtaghi Takyar⁵, Susan Shea⁶ and Kagya A. Amoako^{1*}

¹Department of Chemistry and Chemical & Biomedical Engineering, University of New Haven, West Haven, CT, United States, ²Department of Pediatrics, Yale University, New Haven, CT, United States, ³Analytical Chemistry, Connecticut Agricultural Experiment Station, New Haven, CT, United States, ⁴Yale Cardiovascular Research Center and Yale Cooperative Center of Excellence in Hematology, Yale School of Medicine, New Haven, CT, United States, ⁵Critical Care & Sleep Medicine, Yale University Pulmonary, New Haven, CT, United States, ⁶Department of Surgery, University of Pittsburgh, Pittsburgh, PA, United States

Surface-induced thrombosis remains a significant limitation of blood-contacting medical devices, driven primarily by platelet activation and rapid clot formation. Nitric oxide (NO), a potent endogenous antiplatelet agent, has therefore inspired the development of targeted NO-delivery strategies to mitigate device-associated thrombosis. In this study, nitric oxide-releasing antiplatelet lipid nanoparticles (anti-PLT LNPs) were formulated and characterized for their physicochemical properties and NO-release kinetics. Their platelet aggregation inhibition was evaluated in vitro, while mammalian cell biocompatibility and effects on whole-blood coagulation were evaluated using activated clotting time (ACT) measurements. Anti-PLT LNPs exhibited sustained and measurable NO release after 22 weeks of storage and demonstrated high mammalian cell biocompatibility. The nanoparticles inhibited platelet aggregation by up to 84.4% ($p < 0.01$) and significantly prolonged whole-blood clotting time, resulting in up to a threefold increase in ACT ($p < 0.05$). These findings indicate that NO-releasing anti-PLT LNPs effectively suppress platelet activation and coagulation while maintaining biocompatibility, offering a customizable and targeted approach to mitigate surface-induced thrombosis in blood-contacting medical devices.

KEYWORDS

platelets, liposomes, nanoparticles, blood-contacting medical devices, thrombosis, drug delivery, nitric oxide, anti-coagulation



1 Introduction

Surface-induced thrombosis remains a major challenge in medical devices such as extracorporeal membrane oxygenation (ECMO) circuits, stents, and artificial valves, where altered blood flow and contact with artificial surfaces (AS) initiate protein adsorption, platelet (PLT) activation, and coagulation (1, 2). These reactions increase the risk of device failure, occlusion, and embolization (3–5). Early in AS-induced thrombosis, factor XII (FXII) activation plays a central role, and its deficiency reduces thrombosis without increasing bleeding, making it a compelling therapeutic target (6–9). Clots on the AS form around activated platelets, which provide procoagulant surfaces and support fibrin formation (10–12). Although hydrophilic and zwitterionic coatings such as poly (carboxybetaine methacrylate) (pCBMA), poly (sulfobetaine methacrylate) (pSBMA), and poly (2-methacryloyloxyethyl phosphorylcholine) (pMPC) can reduce protein and platelet adsorption, their application to complex geometries such as tightly packed hollow-fiber oxygenators remains limited by incomplete surface coverage. Platelets are also activated by complement proteins, further amplifying AS thrombosis and emphasizing the value of interventions that act on platelet-driven pathways while minimizing effects on systemic coagulation (9, 13, 14).

Heparin remains the clinical standard for preventing device-associated clotting, acting by enhancing antithrombin III to inhibit factor Xa and thrombin. However, because it blocks both intrinsic (device-mediated) and extrinsic (tissue-mediated) pathways, heparin significantly increases bleeding risk (15, 16). Although selective intrinsic-pathway anticoagulants show

promise (17–21), none are yet device-specific or clinically validated for patients reliant on extracorporeal technologies.

Precision drug delivery offers an alternative strategy by localizing antithrombotic activity while reducing systemic bleeding. Clot-targeted thrombin-cleavable nanoparticles (CTNPs) exemplify this approach by binding activated platelets and fibrin, carrying plasmin, and releasing them in response to thrombin (22, 23). These and other advanced drug-delivery systems (ADDs), including nanocarriers and biomimetic vehicles, aim to improve antithrombotic pharmacokinetics and focus therapy at the thrombus site (23, 24), although translation to transient, device-specific thromboprophylaxis remains limited.

Nitric oxide (NO)-releasing nanomaterials offer another promising pathway. NO regulates vascular tone and inhibits platelet activation, yet its extremely short half-life necessitates controlled-release carriers such as polymeric or inorganic nanoparticles for cardiovascular applications (25). Macromolecular scaffolds are also under investigation to improve NO storage, stability, and targeted delivery (26).

In this study, we investigate liposomal nanoparticles formulated with clinically validated lipids, including those used in COVID-19 vaccines, to evaluate their ability to store and release NO, modulate platelet interactions, assess mammalian cell biocompatibility, and influence whole-blood coagulation. This work aims to establish an active, platelet-focused antithrombotic nanotechnology capable of precision targeting with reduced bleeding risk. Such an approach addresses critical gaps associated with current antithrombotic strategies, including the bleeding complications of systemic anticoagulants, limited validation of FXIIa inhibitors in device-

bearing patients, and the instability or incomplete coverage offered by passive hydrophilic surface coatings.

2 Methods

2.1 Materials

1,2-Distearoyl-sn-glycero-3-phosphocholine (DSPC) and 1,2-Distearoyl-sn-glycero-3-phosphoethanolamine-N-[amino (polyethylene glycol-2000) maleimide (DSPE-PEG maleimide) lipids were purchased from Avanti lipids (Alabaster, AL, USA). Pooled male blood plasma was purchased from ZenBio (Durham, NC, USA), and centrifugations were done using a Beckman Coulter Allegra X-30R centrifuge (Indianapolis, IN, USA). Adenosine diphosphate (ADP), phosphate-buffered saline (PBS), and 1M hydrochloric acid were purchased from Sigma-Aldrich. Diazeniumdiolate dimethylhexane diamine (DMHD/ N_2O_2) NONOate was used as a NO donor. The WI-38 human lung fibroblast cell line was purchased from ATCC (Manassas, VA, USA). WI-38 cells were maintained in minimum essential media (MEM) media (Sigma-Aldrich, Saint Louis, MO, USA) supplemented with 10% fetal bovine serum (FBS) (Gemini Biosciences, Denver, CO, USA), 1% sodium pyruvate, 1% l-glutamine, and 1% penicillin/streptomycin. The cells were maintained at 37 °C, 5% CO_2 . Freshly drawn never frozen bovine blood was purchased from Lampire Biological Laboratories (Everett, PA, USA).

2.2 LNP formulation

The liposome was formulated using 1,2-Distearoyl-sn-glycero-3-phosphocholine and 1,2-Distearoyl-sn-glycero-3-phosphoethanolamine-N-[amino (polyethylene glycol-2000) maleimide lipids (9:1 molar ratio). The process of loading NO into the liposome was performed via the standard thin film hydration method as previously published (27). Briefly, a 2.6 mL NO donor (5, 27) solution [1 mg/mL NO donor in PBS: isopropyl alcohol (9:1 v/v)] was added to a thin lipid film for NO donor encapsulation during liposome formation. A volume of 2 mL of the resulting solution was then suspended in 300 mL of DI water and recirculated in a custom tangential flow filtration system to remove a free NO donor. Encapsulation efficiency (EE) was calculated as the total released NO from a filtered lipid nanoparticle (LNP) solution over that released from a neat NO donor solution $\times 100$. The purified LNPs were stored at 4 °C until they were used in further experiments.

2.3 LNP characterization

Liposomes were imaged by negative staining transmission electron microscopy (Tecnai T12 (Thermo Fisher Scientific, Hillsboro Oregon, USA), 120 kV). Liposome aliquots (4.5 μL , 2 $\times$) were adsorbed to a carbon film grid, and then 4.5 μL of heavy metal salt uranyl acetate was added. The grid was quickly air-dried during

which the heavy metal salt formed an amorphous film embedding the liposomes and imaged to observe liposomes. A liposome solution (100 μL) was added to a quartz cuvette and inserted into the measurement chamber of a Malvern Zetasizer Ultra (The zetasizer at CAES - Malvern Analytical, Westborough, MA, USA). The scattering angle was set to 90° and four runs of measurements per sample were conducted to ensure reproducibility of the collected hydrodynamic diameter and polydispersity index (PDI) data.

2.4 Drug release characterization

NO release measurements were conducted by chemiluminescence using a GE 280i NO Analyzer (NOA) as previously published (28). Briefly, real-time NO release from NO-loaded and unloaded liposome samples was quantified by the chemiluminescence method using a GE 280i NOA, GE Instruments, CO. To measure NO release, 100 μL liposome solutions were injected into a 10 mL PBS (pH 5.5 or 7.34) at a temperature of 37 °C inside the NOA reaction vessel. The acidic pH environment effect on liposomal drug release was of interest as injured thrombotic vascular regions rapidly acidify to create an acidic microenvironment to support healing, influencing factors like angiogenesis, protease activity, and immune cell function. Under such acidic environments, NO release from carriers is typically accelerated. Nitrogen was used as a carrier gas to transport NO from the reaction vessel to the chemiluminescence detection chamber, and NO released from the samples was measured as the concentration detected by NOA (ppb or ppm). NO flux from the samples was calculated as the quotient of NO concentration detected by NOA (ppb or ppm) and the NOA calibration factor ($\text{mol/ppb} \times \text{sec}$), and the calculated surface area (cm^2) of liposomes per 100 μL sample volume and duration NO detection (s or min).

2.5 Platelet aggregation experiment

Pooled human platelet rich plasma (PRP) (3 mL/well) was incubated with equal amounts of PLT activator adenosine diphosphate or fibrinogen-grafted LNPs (w and w/o NO payload) or incubated alone at $n = 4$ replicates in a 24-well plate at 37 °C with gentle agitation for an hour. To ensure thorough mixing, the solution was aspirated and dispensed 6 $\times$. Each group was incubated in triplicate. Three 10 μL volumes from each well were then placed in a row on a glass slide and covered with glass slips. Each sandwiched 10 μL volume was imaged using a Flouid Imager for ImageJ analysis.

2.6 Measurement of platelet aggregation size using ImageJ

An aggregation particle size analysis was performed using ImageJ software (version 1.54g). Fluorescent microscopy images of the particles from all wells ($n = 4$ wells per incubation condition) were imported into ImageJ for the analysis. The measurement process involved image preprocessing where the images were converted to

gray scale and adjusted for brightness and contrast to enhance the clarity of the nanoparticle boundaries. The image scale was set by using the scale bar embedded in the optical image. The “Set Scale” option in ImageJ was used, and the scale bar length and the corresponding real-world measurement in micrometers (100 μm) were entered. The “Threshold” tool was applied to distinguish the particles from the background. A suitable thresholding method (manual adjustment) was selected to isolate particles clearly. The “Analyze Particles” function was used to measure particle size. Parameters such as size range (30–infinity microns²) and circularity (0.1–5) were set to include only particles of interest (ranging from 183 to 679 particles) and exclude artifacts or debris. The “Display Results” and “Show Outlines” options were selected to visualize the identified particles. The software calculated the particle size (area or equivalent diameter) for each identified nanoparticle. The data were exported to a spreadsheet for further statistical analysis. Measurements were reported as mean particle size $\pm$ standard deviation. The accuracy of the particle size measurements was validated by repeating the process with multiple images from independent preparations. This method ensured reliable and reproducible measurements of particle size, enabling quantitative comparisons between different formulations.

2.7 Biocompatibility study

WI-38 cells were maintained in MEM media supplemented with 10% FBS, 1% sodium pyruvate, 1% l-glutamine, and 1% penicillin/streptomycin. The cells were maintained in a 12-well plate at 37 °C, 5% CO₂ after adding anti-PLT LNPs at 50% or 100% MEM media volume in triplicate. Control wells received no anti-PLT LNP. The wells were imaged with the Fluid imager and the cells were counted in Image J to quantify LNP effects. A similar Image J image analysis as described above was followed, except that a particle size range of 10–15 μm was applied. Counts were recorded for further analysis.

2.8 Whole-blood anticoagulation

After recalcification of the whole blood to normal activated clotting time (ACT) and activated partial thromboplastin time (aPTT) values, a static incubation test for up to 10 min of mixing whole blood with LNPs was performed in triplicate by adding 2% of unloaded LNPs and anti-PLT LNPs each to whole bovine blood (v/v). The anticoagulation effect of the anti-PLT LNPs was also evaluated at 10% v/v ($n = 4$ samples). ACT and aPTT data were recorded using the Hemochron Response (Soma Tech. Int., Bloomfield, CT, USA) and Siemens Sysmex CA-560 system Siemens, Malvern, PA, USA.

2.9 Statistics

Data were expressed as mean $\pm$ SD (standard deviation of the mean). The results were analyzed by comparing means using one-

way ANOVA. Values of $p < 0.05$ were considered statistically significant for all tests.

3 Results

3.1 Overview and rationale

Thrombosis on artificial surfaces remains a major clinical challenge due to its association with significant morbidity and mortality. Platelet adhesion and activation initiate thrombus formation and drive its propagation through the release of procoagulant proteins and the generation of fibrin on the activated platelet surface. Because platelets play a central role in thrombus maturation, they represent a crucial therapeutic target. This biological rationale supports the development of anti-PLT LNPs designed to mitigate thrombosis on artificial surfaces by delivering NO, a well-characterized inhibitor of platelet activation.

3.2 Physicochemical characterization of anti-PLT LNPs

3.2.1 Structural design and composition

The design of the NO-loaded anti-PLT LNPs is shown in [Figure 1](#) and incorporates neutral phospholipids, which promote structural stability, effective NO encapsulation, and an antifouling surface, as well as PEG lipids that enhance biocompatibility and allow for controlled NO release. The selection of DSPC and DSPE-PEG maleimide reduces hepatic uptake and renal clearance, supporting prolonged circulation, a key requirement for thrombosis-prevention strategies.

3.2.2 Particle size, zeta potential, and homogeneity

Characterization using transmission electron microscope (TEM), dynamic light scattering (DLS), and NO analysis demonstrated that the LNPs possessed a zeta potential of -29.45 ± 0.07 mV, a polydispersity index of 0.41, and an average diameter of 62.52 ± 9.47 nm ([Figures 1A–C](#)). These values indicate a stable and homogeneous nanoparticle suspension with a physiologically favorable size and surface characteristics. Such properties are consistent with reduced macrophage clearance and extended blood residence times, as reported in previous studies ([29–33](#)).

3.2.3 NO encapsulation and release profile

An assessment of NO release from 100 μL aliquots of LNPs in PBS at physiological pH (7.34) and acidic pH (5.5) at 37 °C revealed an entrapment efficiency of 49.87% ([Figures 1D,E](#)). The NO flux was 79.8×10^{-5} mol/min cm^2 under physiological conditions and increased to 598.4×10^{-5} moles/min cm^2 under acidic conditions, demonstrating a pH-responsive release mechanism. This controlled, stimulus-dependent release profile confirms that the LNPs retain functional NO payload capacity under physiologically relevant conditions and provide sustained antiplatelet activity over time.

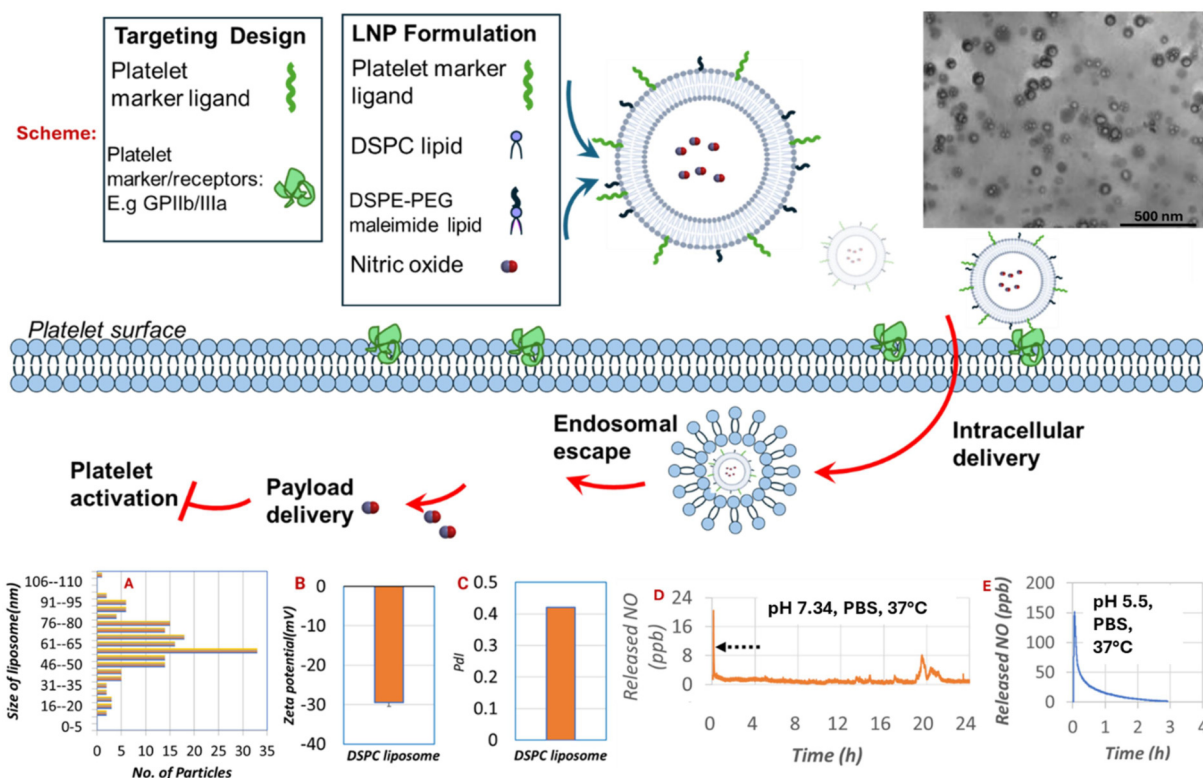


FIGURE 1

Physicochemical characterization of neutral/PEG LNPs loaded with an antiplatelet drug payload and functionalized with fibrinogen ligand. Design and drug-delivery principles are shown along with a TEM image of the LNPs. The panels include size distribution (A), zeta potential (B), PDI (C), and nitric oxide release profiles under physiological blood pH (D) and acidic pH (E) conditions.

3.3 Storage stability of LNP formulations

3.3.1 Relevance of long-term stability

Long-term storage stability is critical for the clinical deployment of LNP-based therapeutics. Stable formulations maintain structural integrity, encapsulation efficiency, and functional drug release profiles during storage, which are essential for consistent therapeutic performance, large-scale manufacturing, and distribution.

3.3.2 Fresh vs. stored LNP functionality

Comparative NO release analyses revealed that freshly prepared LNPs exhibited approximately threefold higher total NO release compared with LNPs stored at 4 °C ($p < 0.05$) (Figure 2). This suggests the gradual loss of encapsulated NO during storage, likely due to physicochemical instability or increased membrane permeability over time. While the primary objective of the study did not include conducting a full temporal stability assessment, such temporal profiling would provide additional insights.

3.3.3 Strategies to improve storage stability

Potential strategies to enhance stability include the incorporation of cholesterol, which increases membrane rigidity and reduces drug leakage (34), and the application of

freeze-drying procedures with cryoprotectants such as dextrose or lactose, which preserve structural integrity and prevent aggregation during dehydration and rehydration (35). Further studies assessing size dynamics, membrane integrity, and functional NO release across storage conditions are warranted to optimize long-term stability.

3.4 Antiplatelet activity of NO-loaded LNPs

3.4.1 Inhibition of platelet aggregation

The antiplatelet efficacy of the NO-loaded LNPs was assessed by comparing their effects on platelet aggregation with untreated controls and known platelet activators (Figure 3). LNP-treated platelet samples displayed a marked reduction in aggregate formation. Microscopy images corroborated the quantitative data, and ANOVA confirmed statistically significant differences between treatment groups.

3.4.2 Aggregate surface coverage and morphology

Platelets in the untreated control exhibited the most extensive overall aggregation, although the resulting aggregates did not form large clumps. In comparison, the LNP shell, LNP with fibrinogen, and ADP groups developed larger aggregates, with

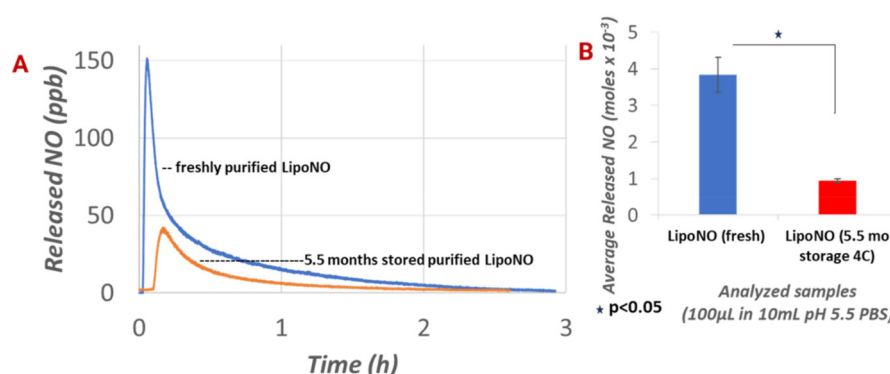


FIGURE 2

NO release from anti-PLT LNPs measured using GE 280i NOA. Nitric oxide release was measured from a 100 μ L of anti-PLT LNP solution reconstituted in a 10 mL PBS at pH 5.5. Freshly purified LipoNO exhibited a higher NO release peak compared with LipoNO stored at 4 $^{\circ}$ C for over 5 months (A) The average total moles of NO released ($n = 3$ per group) also differed significantly between the fresh and the stored formulations (B) NOA, nitric oxide analyzer; PBS, phosphate-buffered saline; PLT, platelet.

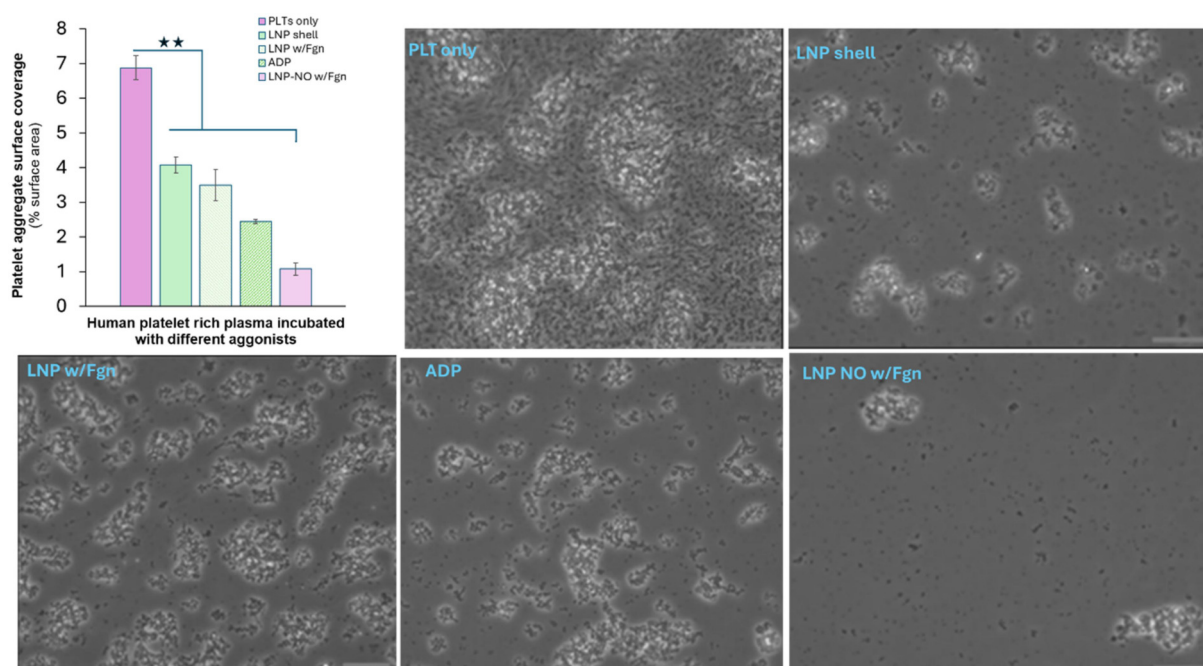


FIGURE 3

Platelet aggregation in human platelet-rich plasma after 1 h incubation. Platelet-rich plasma was incubated for 1 h in a 96-well plate (37 $^{\circ}$ C, 35 RPM) under five conditions: no agonist (PLT only), empty liposomes (LNP shell), fibrinogen-grafted LNPs (LNP w/Fgn), ADP, and nitric oxide-loaded, fibrinogen-grafted LNPs (LNP NO w/Fgn). Platelet aggregation was quantified as the percentage of surface area covered by platelet macroaggregates. Scale bar: 100 μ m.

the fibrinogen-containing LNPs producing the greatest aggregate surface area. In contrast, the NO-loaded LNPs with fibrinogen produced the smallest aggregate surface coverage, consistent with effective NO-mediated inhibition of platelet activation. Although additional platelet identity and activation markers (CD41, CD62P) were not evaluated in this study, future analyses incorporating these markers could provide further insight into platelet activation status.

3.5 Biocompatibility of anti-PLT LNPs

3.5.1 Importance of biocompatibility assessment

Evaluating LNP biocompatibility is essential to ensure safety in mammalian cell environments and to minimize cytotoxicity or adverse cellular responses. This is particularly important for NO-releasing formulations intended for prolonged blood-contact applications.

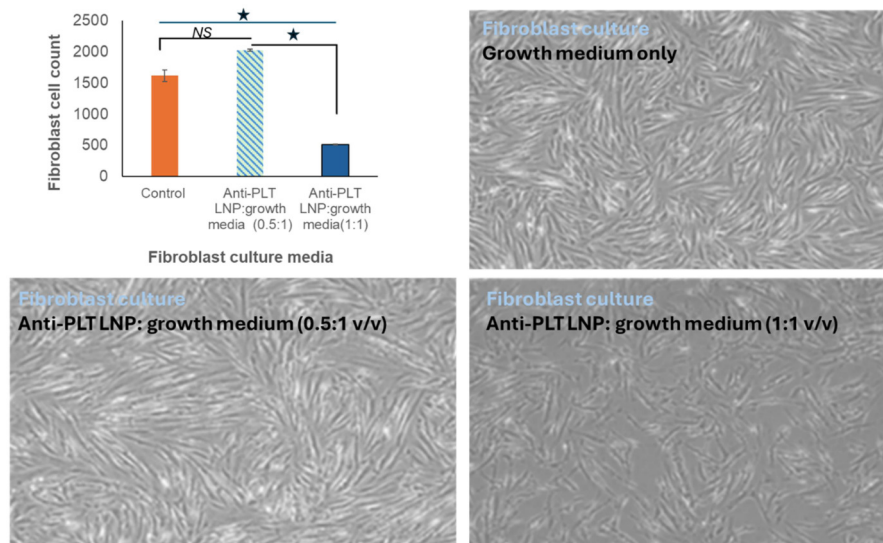


FIGURE 4

Biocompatibility of anti-PLT LNPs. Light microscopy images of human fibroblasts cultured for 24 h with anti-PLT LNPs mixed with a growth medium at ratios of 0:1 (control), 0.5:1 ($9.60 \pm 0.09 \times 10^{-3}$ mol NO exposure), and 1:1 ($19.20 \pm 3.68 \times 10^{-3}$ mol NO exposure) show that cell population density in the 0.5:1 group was comparable to that in the control group. In contrast, a reduction in cell density was observed at the 1:1 ratio, consistent with the known antiproliferative effects of higher NO concentrations.

3.5.2 Cell viability outcomes

Fibroblast viability data shown in Figure 4 indicate that low-dose LNP treatment (0.5:1 LNP:medium) did not induce cytotoxicity and resulted in cell counts that exceeded control levels, likely reflecting low-concentration NO-mediated proliferative effects described in the literature (36, 37). At higher doses (1:1), the expected antiproliferative effects of elevated NO concentrations were observed (38), yet overall biocompatibility remained within acceptable limits. These findings support the safety of anti-PLT LNPs across a broad range of therapeutically relevant concentrations.

3.6 Effects on blood coagulation parameters

3.6.1 Modulation of ACT and aPTT

The anticoagulant potential of NO-loaded, fibrinogen-grafted LNPs was evaluated by measuring ACT and aPTT in whole blood (Figure 5). The addition of 2% LNPs significantly increased both ACT and aPTT, indicating effective delay of clot initiation. Increasing the LNP concentration to 10% produced a dose-dependent prolongation of ACT, with sustained effects observed even 300 s after LNP addition.

3.6.2 Implications for blood-contacting medical devices

These findings demonstrate that anti-PLT LNPs exert both antiplatelet and anticoagulant effects and may therefore improve the performance of artificial surfaces used in blood-contacting devices. By reducing thrombosis formation on oxygenators,

hemodialysis circuits, and other extracorporeal technologies, these LNPs have the potential to lower device-related failure rates and enhance patient outcomes in clinical settings.

4 Discussion

The present study demonstrates the successful formulation, characterization, and functional evaluation of nitric oxide-releasing antiplatelet lipid nanoparticles designed to mitigate thrombosis on artificial surfaces. These findings address a critical need in the field of blood-contacting medical devices, where thrombus formation remains a major barrier to long-term device performance. Platelet-mediated thrombosis is well recognized as the primary initiator of clot formation on biomaterials, driven by rapid platelet adhesion, activation, and aggregation on foreign surfaces (6). Our data align with this established paradigm by showing extensive platelet aggregation in untreated control samples and in groups exposed to prothrombotic stimuli such as fibrinogen or ADP. The strong antiaggregatory effect observed with NO-loaded LNPs reinforces the central role of NO as a potent endogenous inhibitor of platelet activation via cyclic guanosine monophosphate (cGMP)-dependent pathways (39, 40).

The physicochemical properties of the LNPs contributed to their functional performance and therapeutic potential. The observed size, mildly negative surface charge, and moderate PDI are consistent with optimally engineered nanoparticles designed to evade rapid clearance by the mononuclear phagocyte system while maintaining colloidal stability (41, 42). PEGylation has long been known to extend circulation half-life and reduce

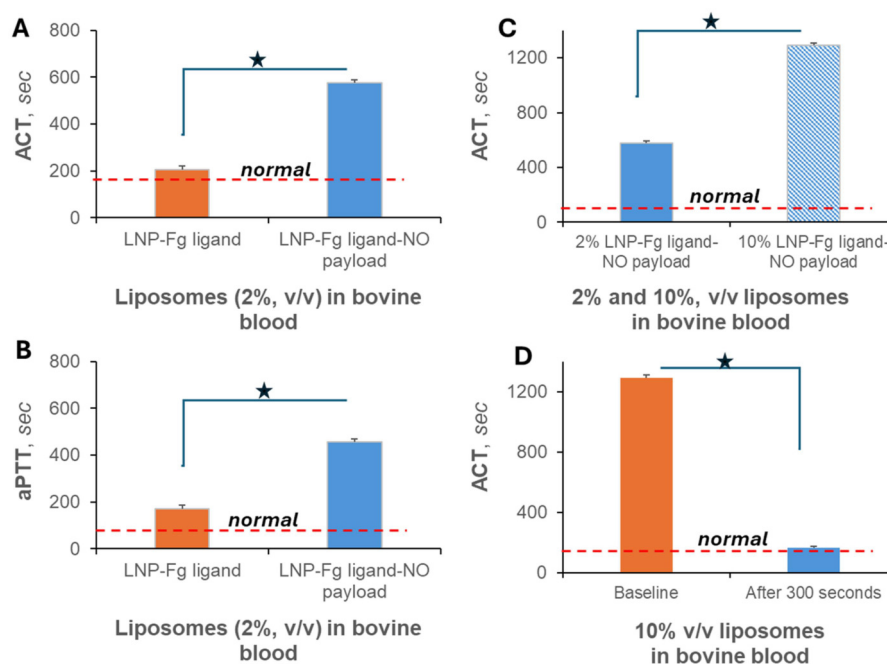


FIGURE 5

Effect of anti-PLT LNPs on whole-blood coagulation markers. ACT (A) and aPTT (B) increased following the addition of 2% NO-loaded, fibrinogen-grafted LNPs (v/v). ACT measurements further demonstrated a dose-dependent response, with higher LNP concentrations (2% vs. 10%) producing greater prolongation 1 min after LNP addition (C). The sustained effect of 10% LNPs on ACT, measured 300 s post addition, is shown in panel (D).

protein adsorption and opsonization (43–45). The design choices made here, specifically the inclusion of DSPC for membrane rigidity and DSPE-PEG for hydrophilicity, align with previous LNP formulations that demonstrate prolonged systemic residence and reduced immunogenicity (46, 47). These structural features likely contributed to the stable NO retention and controlled release observed in our NO flux studies.

The pH-responsive NO release profile further confirms the functional precision of the LNP formulation. Increased NO release at acidic pH is consistent with that of prior studies showing that acid-labile NO donors or acid-sensitive LNP matrices can enhance NO liberation in microenvironments associated with platelet activation or inflamed tissue (48, 49). Physiologically, activated platelets and developing thrombi can produce localized acidic microenvironments, promoting targeted NO release at the site of thrombosis (50). Thus, the observed pH-dependent release behavior provides a mechanistic advantage over conventional systemic NO donors, which suffer from short half-life and off-target effects (51).

The stability studies revealed a reduction in NO release in stored LNPs relative to freshly prepared formulations. This finding is consistent with reports that NO is prone to diffusion or degradation during prolonged storage, especially in lipid bilayers lacking sufficient rigidity or protective excipients (52). Our results align with prior work showing improved stability when cholesterol is incorporated into liposomal membranes to reduce permeability (53, 54) and represents a logical next step to enhance storage stability for translational applications.

Functional assays provided compelling evidence of the therapeutic potential of the NO-loaded LNPs. Platelet aggregation studies demonstrated significant inhibition of platelet clumping, consistent with earlier reports that LNP-mediated NO delivery can attenuate platelet activation by suppressing intracellular calcium mobilization and integrin $\alpha\text{IIb}\beta 3$ activation (55, 56). The pronounced reduction in aggregate surface area in the NO-LNP groups compared with fibrinogen- or ADP-stimulated controls mirrors findings from studies in which NO donors prevent the transition from platelet adhesion to stable aggregation (57, 58). Although platelet activation markers such as CD41 and CD62P were not assessed in this study, their inclusion in future work would strengthen mechanistic conclusions.

Biocompatibility results demonstrated that anti-PLT LNPs are safe across a range of concentrations, with low-dose formulations even promoting fibroblast proliferation. This is consistent with evidence that low-level NO stimulates cell growth, migration, and collagen synthesis (59, 60). Conversely, the antiproliferative effect observed at higher concentrations aligns with the well-documented biphasic role of NO, in which elevated levels inhibit mitochondrial respiration and induce cytostasis (61, 62). These findings confirm that the LNP-delivered NO behaves within known physiological boundaries and supports the safety of the formulation for eventual *in vivo* use.

The anticoagulant effects observed in whole blood further strengthen the translational relevance of the system. Prolonged ACT and aPTT values following treatment with NO-loaded,

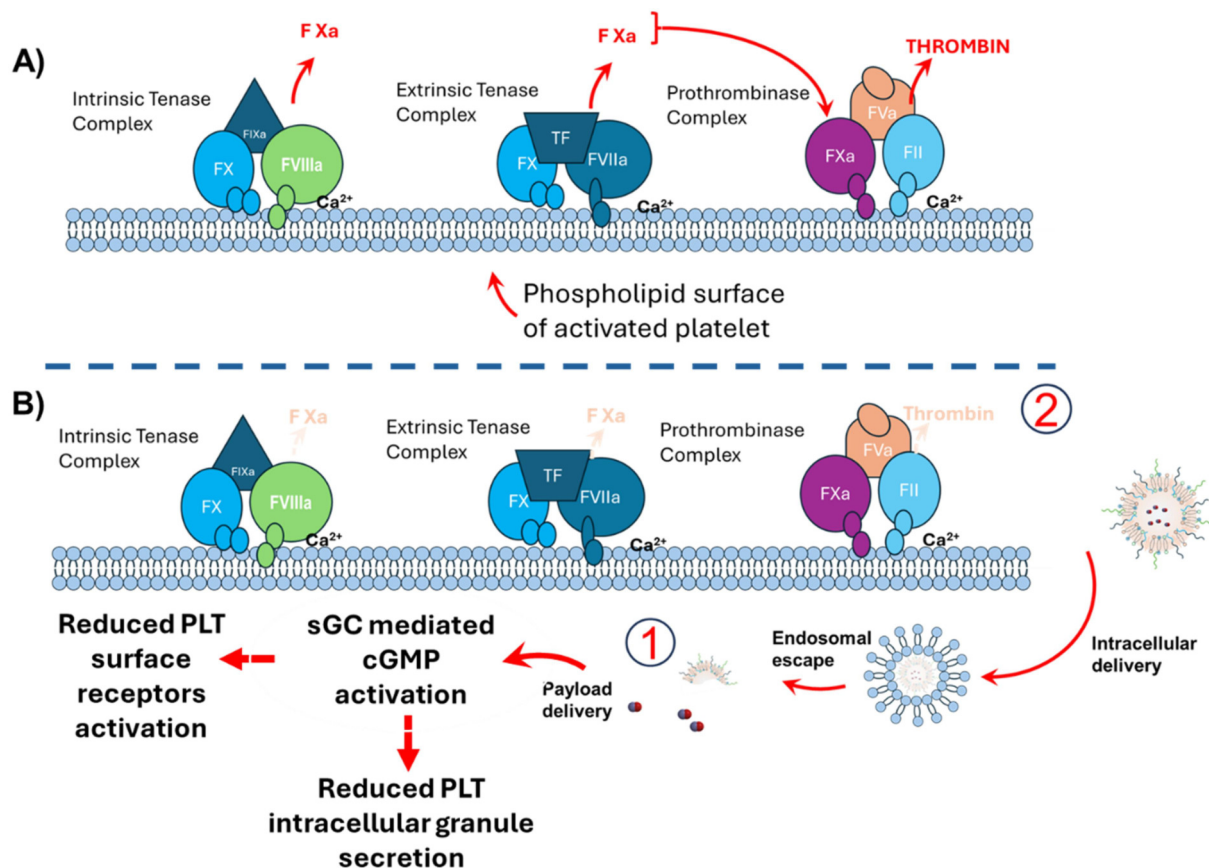


FIGURE 6

Proposed mechanism of action of anti-PLT LNPs. Clot propagation relies on the activation of factor X and the generation of thrombin on the phospholipid surface of activated platelets (A), where assembly of the tenase and prothrombinase complexes occurs. Delivery of NO from anti-PLT LNPs is proposed to inhibit platelet activation through soluble guanylyl cyclase (sGC)-mediated cGMP activation, thereby limiting the formation of these coagulation complexes and subsequent fibrin production (B). As platelet activation reduces, the assembly of procoagulant complexes on the platelet surface diminishes, ultimately leading to decreased clot formation.

fibrinogen-grafted LNPs are consistent with NO-mediated inhibition of thrombin generation and reduced fibrin formation, as previously described in NO-based anticoagulant research (63, 64). The dose-dependent and sustained prolongation of ACT at higher LNP concentrations suggests that the system is capable of producing long-lasting anticoagulation, an essential property for application in extracorporeal circuits, vascular grafts, and artificial lungs, all of which are highly prone to thrombosis (65, 66).

We illustrate the proposed mechanism of action for anti-PLT LNPs in reducing coagulation (Figure 6). During clot formation, the activation of clotting factor X and the release of thrombin from platelets are supported by the phospholipid surface of activated platelets, which assembles the protein complexes necessary for these processes to occur (top panel). Since the anti-PLT LNPs were designed to deliver nitric oxide to inhibit platelet activation, it is thought that a reduction in platelet activation will reduce the assembly of the tenase and prothrombinase complexes, as well as fibrin formation on the platelet surface (bottom panel). A diminished activation is expected to disrupt the cascade of events leading to clot propagation, resulting in reduced thrombus formation.

This mechanistic insight underscores the potential of anti-PLT LNPs to effectively target and mitigate platelet-driven thrombotic processes.

5 Conclusion

Taken together, the results of this study provide compelling evidence that NO-loaded anti-PLT LNPs represent a promising strategy for reducing thrombosis on artificial surfaces. Their physicochemical stability, controlled NO release, strong antiplatelet effects, and favorable biocompatibility profile position them as a viable platform for translation into clinical settings. Despite these promising findings, the study is limited by its reliance on *in vitro* models and the absence of long-term stability and mechanistic assessment gaps (66–73). In addition, manufacturability and scalability of the anti-PLT LNPs would require high throughput and scalable formulation technology. Future work will focus on optimizing formulation stability (e.g., cholesterol incorporation and cryoprotective lyophilization), integrating these LNPs into extracorporeal membrane

oxygenation or stent coatings for dynamic flow testing, validating efficacy under dynamic flow and *in vivo* thrombosis models, incorporating mechanistic platelet activation assays, and studying rapid formulation technologies. With continued refinement, these LNPs could significantly improve the hemocompatibility and functional lifetime of a broad range of medical devices, optimizing long-term storage stability.

Data availability statement

The datasets presented in this article are not publicly available due to commercial confidentiality and proprietary restrictions. Requests to access the data should be directed to corresponding author, subject to approval by the data owner.

Ethics statement

Ethical approval was not required for the studies on animals in accordance with the local legislation and institutional requirements because only commercially available established cell lines were used.

Author contributions

SM: Data curation, Formal analysis, Investigation, Methodology, Validation, Writing – review & editing. AM: Data curation, Investigation, Methodology, Validation, Writing – review & editing. PP: Investigation, Methodology, Writing – review & editing. VF: Investigation, Methodology, Writing – review & editing. NZ-M: Data curation, Methodology, Writing – review & editing. JH: Investigation, Methodology, Supervision, Writing – review & editing. ST: Investigation, Methodology, Supervision, Writing – review & editing. SS: Methodology, Writing – review & editing. KA: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

References

- Centers for Disease Control and Prevention. *Chronic Kidney Disease in the United States, 2021*. Atlanta, GA: US Department of Health and Human Services, Centers for Disease Control and Prevention (2021).
- National Center for Health Statistics. *Percentage of Angina for Adults Aged 18 and Over*. Hyattsville, MD: National Center for Health Statistics, Centers for Disease Control and Prevention (2020). p. 2019–20.
- Amoako KA. *Nitric oxide therapies for local inhibition of platelets' activation on blood-contacting surfaces* (PhD dissertation). Ann Arbor, MI: The University of Michigan (2011).
- Gbyli R, Amoako KA. "Improving the hemocompatibility of biomedical polymers". In: Siedlecki C, editor. *Hemocompatibility of Biomaterials for Clinical Applications*. Cambridge: Woodhead Publishing (Elsevier) (2018). p. 223–52.
- Gbyli R, Zito C, Amoako KA. *In vitro* cytocompatibility of antibacterial levels of polymer nitric oxide release. *Eng Press*. (2018) 2(1):66–72. doi: 10.28964/EngPress-2-113
- Gorbet MB, Sefton MV. Biomaterial-associated thrombosis: roles of coagulation factors, complement, platelets and leukocytes. *Biomaterials*. (2004) 25(26):5681–703. doi: 10.1016/j.biomaterials.2004.01.023

Funding

The author(s) declared that financial support was received for this work and/or its publication. This work was supported in part by internal funding from the Biomedical Engineering program at the University of New Haven. The funders had no role in the study design; collection, analysis, or interpretation of data; writing of the manuscript; or the decision to submit it for publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial intelligence, and reasonable efforts have been made to ensure accuracy, including review by the authors, wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fmedt.2025.1690389/full#supplementary-material>

7. Anderson JM, Rodriguez A, Chang DT. Foreign body reaction to biomaterials. *Semin Immunol.* (2008) 20(2):86–100. doi: 10.1016/j.smim.2007.11.004
8. Mair GE, Olson SR, Shatzel JJ. The safety of novel drugs targeting factor XI and XII in human clinical trials: a systematic review. *Blood.* (2019) 134(Supplement_1):4966. doi: 10.1182/blood-2019-130823
9. Pham M, Stoll G, Nieswandt B, Bendszus M, Kleinschnitz C. Blood coagulation factor XII—a neglected player in stroke pathophysiology. *J Mol Med.* (2012) 90:119–26. doi: 10.1007/s00109-011-0812-9
10. Kakali G, Maitree B. Overview of platelet physiology: its hemostatic and nonhemostatic role in disease pathogenesis. *Sci World J.* (2014) 2014:781857. doi: 10.1155/2014/781857
11. Butenas S, Orfeo T, Mann KG. Tissue factor activity and function in blood coagulation. *Thromb Res.* (2008) 122(Suppl 1):S42–6. doi: 10.1016/S0049-3848(08)70018-5
12. Krishnaswamy S, Nesheim ME, Prydzial EL, Mann KG. Assembly of prothrombinase complex. *Methods Enzymol.* (1993) 222:260–80. doi: 10.1016/0076-6879(93)222018-B
13. Subramaniam S, Jurk K, Hobohm L, Jäckel S, Saffarzadeh M, Schwierczek K, et al. Distinct contributions of complement factors to platelet activation and fibrin formation in venous thrombus development. *Blood.* (2017) 129(16):2291–302. doi: 10.1182/blood-2016-11-749879
14. Foley JH, Conway EM. Cross talk pathways between coagulation and inflammation. *Circ Res.* (2016) 118(9):1392–408. doi: 10.1161/CIRCRESAHA.115.306578
15. Dalton HJ, Garcia-Filion P, Holubkov R, Moler FW, Shanley T, Heidemann S, et al. Association of bleeding and thrombosis with outcome in extracorporeal life support. *Pediatr Crit Care Med.* (2015) 16(2):167–74. doi: 10.1097/PCC.0000000000000317
16. Mazzeffi M, Greenwood J, Tanaka K, Menaker J, Rector R, Herr D, et al. Bleeding, transfusion, and mortality on extracorporeal life support: ECLS working group on thrombosis and hemostasis. *Ann Thoracic Surg.* (2016) 101(2):682–9. doi: 10.1016/j.athoracsurg.2015.07.046
17. Wilbs J, Kong XD, Middendorp SJ, Prince R, Cooke A, Demarest CT, et al. Cyclic peptide FXII inhibitor provides safe anticoagulation in a thrombosis model and in artificial lungs. *Nat Commun.* (2020) 11(1):3890. doi: 10.1038/s41467-020-17648-w
18. Naito N, Ukita R, Wilbs J, Wu K, Lin X, Carleton NM, et al. Combination of polycarboxybetaine coating and factor XII inhibitor reduces clot formation while preserving normal tissue coagulation during extracorporeal life support. *Biomaterials.* (2021) 272:120778. doi: 10.1016/j.biomaterials.2021.120778
19. Weitz JI, Eikelboom JW. What is the future of factor XI inhibitors? *Circulation.* (2022) 146(25):1899–902. doi: 10.1161/CIRCULATIONAHA.122.061132
20. Presume J, Ferreira J, Ribeiras R. Factor XI inhibitors: a new horizon in anticoagulation therapy. *Cardiol Ther.* (2024) 13(1):1–16. doi: 10.1007/s40119-024-00352-x
21. Umei N, Shin S, Lai A, Miller J, Roberts K, Strelkova D, et al. Factor XII silencing using siRNA prevents thrombus formation in a rat model of extracorporeal life support. *ASAIO J.* (2023) 69(6):527–32. doi: 10.1097/MAT.0000000000001876
22. Sun M, Hao Pontius MH, Yang S, Pendekanti T, Raghunathan S, Shavit JA, et al. Direct delivery of plasmin using clot-anchoring thrombin-responsive nanoparticles for targeted fibrinolytic therapy. *J Thromb Haemost.* (2023) 21(4):983–94. doi: 10.1016/j.jtha.2022.11.037
23. Greineder CF, Howard MD, Carnemolla R, Cines DB, Muzykantov VR. Advanced drug delivery systems for antithrombotic agents. *Blood.* (2013) 122(9):1565–75. doi: 10.1182/blood-2013-03-453498
24. Absar S, Kwon YM, Ahsan F. Bio-responsive delivery of tissue plasminogen activator for localized thrombolysis. *J Control Release.* (2014) 177:42–50. doi: 10.1016/j.jconrel.2013.12.036
25. Tabish TA, Crabtree MJ, Townley HE, Winyard PG, Lygate CA. Nitric oxide releasing nanomaterials for cardiovascular applications. *JACC Basic Transl Sci.* (2023) 9(5):691–709. doi: 10.1016/j.jacbs.2023.07.017
26. Riccio DA, Schoenfish MH. Nitric oxide release: part I. macromolecular scaffolds. *Chem Soc Rev.* (2012) 41(10):3731–41. doi: 10.1039/c2cs35058a
27. Amoako K, Kaufman R, Haddad WAM, Pusey R, Saniesetty VHK, Sun H, et al. Zwitterionic polysulfobetaine coating and antiplatelet liposomes reduce fouling in artificial lung circuits. *Macromol Biosci.* (2023) 23:2200479. doi: 10.1002/mabi.202200479
28. Gupta S, Amoako KA, Suhaib A, Cook KE. Multi-modal, surface-focused anticoagulation using poly-2-methoxyethylacrylate polymer grafts and surface nitric oxide release. *Adv Mater Interfaces.* (2014) 1:1400012. doi: 10.1002/admi.201400012
29. Liu Q, Guan J, Qin L, Zhang X, Mao S. Physicochemical properties affecting the fate of nanoparticles in pulmonary drug delivery. *Drug Discov Today.* (2020) 25:150–9. doi: 10.1016/j.drudis.2019.10.003
30. Blackman LD, Gunatillake PA, Cass P, Locock KES. An introduction to zwitterionic polymer behavior and applications in solution and at surfaces. *Chem Soc Rev.* (2019) 48:757–70. doi: 10.1039/c8cs00864f
31. Tsao C, Yuan Z, Zhang P, Liu E, McMullen P, Wu K, et al. Enhanced pulmonary systemic delivery of protein drugs via zwitterionic polymer conjugation. *J Control Release.* (2020) 322:170–6. doi: 10.1016/j.jconrel.2020.03.021
32. Samuelsson E, Shen H, Blanco E, Ferrari M, Wolfram J. Contribution of Kupffer cells to liposome accumulation in the liver. *Colloids Surf B Biointerfaces.* (2017) 158:356–62. doi: 10.1016/j.colsurfb.2017.07.014
33. Miller MR, Raftis JB, Langrish JP, McLean SG, Samutrtai P, Connell SP, et al. Inhaled nanoparticles accumulate at sites of vascular disease. *ACS Nano.* (2017) 11(5):4542–52. doi: 10.1021/acsnano.6b08551
34. Kaddah S, Khreich N, Kaddah F, Charcosset C, Greige-Gerges H. Cholesterol modulates the liposome membrane fluidity and permeability for a hydrophilic molecule. *Food Chem Toxicol.* (2018) 113:40–8. doi: 10.1016/j.fct.2018.01.017
35. Boafu GF, Magar KT, Ekpo MD, Qian W, Tan S, Chen C. The role of cryoprotective agents in liposome stabilization and preservation. *Int J Mol Sci.* (2022) 23(20):12487. doi: 10.3390/ijms232012487
36. Napoli C, Paolisso G, Casamassimi A, Al-Omran M, Barbieri M, Sommese L, et al. Effects of nitric oxide on cell proliferation: novel insights. *J Am Coll Cardiol.* (2013) 62(2):89–95. doi: 10.1016/j.jacc.2013.03.070
37. Won JE, Kim WJ, Shim JS, Ryu JJ. Guided bone regeneration with a nitric-oxide releasing polymer inducing angiogenesis and osteogenesis in critical-sized bone defects. *Macromol Biosci.* (2022) 22:2200162. doi: 10.1002/mabi.202200162
38. Zhao N, Tian K, Cheng K, Han T, Hu X, Li D, et al. Antiproliferative activity and apoptosis inducing effects of nitric oxide donating derivatives of evodiamine. *Bioorg Med Chem.* (2016) 24(13):2971–8. doi: 10.1016/j.bmc.2016.05.001
39. Radomski MW, Palmer RM, Moncada S. An L-arginine/nitric oxide pathway present in human platelets regulates aggregation. *Proc Nat Acad Sci U S A.* (1990) 87(13):5193–7. doi: 10.1073/pnas.87.13.5193
40. Loscalzo J, Welch G. Nitric oxide and its role in the cardiovascular system. *Prog Cardiovasc Dis.* (1995) 38(2):87–104. doi: 10.1016/S0033-0620(05)80001-5
41. Zhang X, Li X, Gao X, Hu J, Sun Y. Nanoparticle-based nitric oxide delivery and its therapeutic applications. *J Control Release.* (2013) 172(3):618–28. doi: 10.1016/j.jconrel.2023.08.038
42. Kulkarni JA, Darjuan MM, Mercer JE, Chen S, van der Meel R, Thewalt JL, et al. On the formation and morphology of lipid nanoparticles containing ionizable cationic lipids and siRNA. *ACS Nano.* (2018) 12(5):4787–95. doi: 10.1021/acsnano.8b01516
43. Owens DE, Peppas NA. Opsonization, biodistribution, and pharmacokinetics of polymeric nanoparticles. *Int J Pharm.* (2006) 307(1):93–102. doi: 10.1016/j.ijpharm.2005.10.010
44. Suk JS, Xu Q, Kim N, Hanes J, Ensign LM. Pegylation as a strategy for improving nanoparticle-based drug and gene delivery. *Adv Drug Deliv Rev.* (2016) 99:28–51. doi: 10.1016/j.addr.2015.09.012
45. Koziara JM, Oh JJ, Akers WS, Ferraris SP, Mumper RJ. Blood compatibility of cetyl alcohol/polysorbate-based nanoparticles. *Pharm Res.* (2005) 22(11):1821–8. doi: 10.1007/s11095-005-7547-7
46. Semple SC, Akinc A, Chen J, Sandhu AP, Mui BL, Cho CK, et al. Rational design of cationic lipids for siRNA delivery. *Nat Biotechnol.* (2010) 28(2):172–6. doi: 10.1038/nbt.1602
47. Hou X, Zaks T, Langer R, Dong Y. Lipid nanoparticles for mRNA delivery. *Nat Rev Mater.* (2021) 6(12):1078–94. doi: 10.1038/s41578-021-00358-0
48. Yu H, Cui LX, Huang N, Yang ZL. Recent developments in nitric oxide-releasing biomaterials for biomedical applications. *Med Gas Res.* (2019) 9(4):184–91. doi: 10.4103/2045-9912.273956
49. Carpenter AW, Schoenfish MH. Nitric oxide release: part II. Therapeutic applications. *Chem Soc Rev.* (2012) 41(10):3742–52. doi: 10.1039/c2cs15273h
50. Liu C, Zhang W, Li Y, Chen Y, Leng X, Sun S. Liposome-based nanocarriers for drug delivery: design, preparation and application. *Colloids Surf B Biointerfaces.* (2017) 158:10–20. doi: 10.1016/j.colsurfb.2017.05.012
51. Hwang D, Ramsey JD, Kabanov AV. Polymeric micelles for the delivery of poorly soluble drugs: From nanoformulation to clinical approval. *Adv Drug Deliv Rev.* (2020) 156:80–118. doi: 10.1016/j.addr.2020.09.009
52. Lasic DD, Papahadjopoulos D. Liposomes revisited. *Science.* (1995) 267(5202):1275–6. doi: 10.1126/science.7871422
53. Immordino ML, Dosio F, Cattel L. Stealth liposomes: review of the basic science, rationale, and clinical applications, existing and potential. *Int J Nanomed.* (2006) 1(3):297–315. doi: 10.2217/17435889.1.3.297
54. Bechtold-Peters K, Linden M, Kissel T. Comparison of spray-drying, lyophilization, and vacuum-drying techniques for the stabilization of peptide-loaded poly(lactide-co-glycolide) nanoparticles. *J Control Release.* (2001) 72(1–3):433–45.

55. Major JM, Acevedo MJ, Bicha RM. Stability of liposomes during long-term storage: influence of composition and processing method. *J Pharm Sci.* (2010) 99(3):1253–64.
56. Freedman JE, Loscalzo J, Barnard MR, Alpert C, Michelson AD. Nitric oxide released from activated platelets inhibits platelet recruitment. *J Clin Invest.* (1997) 100(2):350–6. doi: 10.1172/JCI119540
57. Li Y, Jin Y, He X, Tang Y, Zhou M, Guo W, et al. Cyclo(RGD) peptide-decorated silver nanoparticles with anti-platelet potential for active platelet-rich thrombus targeting. *Nanomedicine.* (2022) 41:102520. doi: 10.1016/j.nano.2022.102520
58. Thomas DD, Ridnour LA, Isenberg JS, Flores-Santana W, Switzer CH, Donzelli S, et al. The chemical biology of nitric oxide: implications in cellular signaling. *Free Radic Biol Med.* (2008) 45(1):18–31. doi: 10.1016/j.freeradbiomed.2008.03.020
59. Ridnour LA, Isenberg JS, Espey MG, Thomas DD, Roberts DD, Wink DA. Nitric oxide regulates angiogenesis through a functional switch involving thrombospondin-1. *Proc Nat Acad Sci U S A.* (2004) 102(37):13147–52. doi: 10.1073/pnas.0502979102
60. Murphy MP. Nitric oxide and cell death. *Biochim Biophys Acta.* (1999) 1411(2–3):401–14. doi: 10.1016/S0005-2728(99)00029-8
61. Wink DA, Mitchell JB. Chemical biology of nitric oxide: insights into regulatory, cytotoxic, and cytoprotective mechanisms of nitric oxide. *Free Radic Biol Med.* (1998) 25(4–5):434–56. doi: 10.1016/S0891-5849(98)00092-6
62. Yang C, Hwang HH, Jeong S, Seo D, Jeong Y, Lee DY, et al. Inducing angiogenesis with the controlled release of nitric oxide from biodegradable and biocompatible copolymeric nanoparticles. *Int J Nanomedicine.* (2018) 13:6517–30. doi: 10.2147/IJN.S174989
63. Seabra AB, Durán N, Haddad PS. Nitric oxide-releasing nanomaterials for biomedical applications. *Curr Med Chem.* (2016) 23(25):2724–42. doi: 10.1016/j.niox.2025.03.004
64. Pratt PF, Radomski MW, Palmer RM, Moncada S. Platelet-derived nitric oxide inhibits platelet adhesion to vascular endothelium. *J Cardiovasc Pharmacol.* (1997) 29(2):178–82. doi: 10.1016/s0140-6736(87)91481-4
65. Ren T, Mi Y, Wei J, Han X, Zhang X, Zhu Q, et al. Advances in nano-functional materials in targeted thrombolytic drug delivery. *Molecules.* (2024) 29(10):2325. doi: 10.3390/molecules29102325
66. Datta A, Dasgupta S, Mukherjee S. Modifications of nano-titania surface for *in vitro* evaluations of hemolysis, cytotoxicity, and nonspecific protein binding. *J Nanopart Res.* (2017) 19:142. doi: 10.1007/s11051-017-3835-5
67. Worm M, Köhler EC, Panda R, Long A, Butler LM, Stavrou EX, et al. The factor XIIa blocking antibody 3F7: a safe anticoagulant with anti-inflammatory activities. *Ann Transl Med.* (2015) 3:17. doi: 10.3978/j.issn.2305-5839.2015.07.06
68. Miller CS, Grandi SM, Shimony A, Filion KB, Eisenberg MJ. Meta-analysis of efficacy and safety of new oral anticoagulants (dabigatran, rivaroxaban, apixaban) versus warfarin in patients with atrial fibrillation. *Am J Cardiol.* (2012) 110(3):453–60. doi: 10.1016/j.amjcard.2012.03.049
69. Meredith T, Schnegg B, Hayward C. The use of direct oral anticoagulants in patients with ventricular assist devices: is there hope for factor Xa inhibition? *Artif Organs.* (2021) 45(5):E123–9. doi: 10.1111/aor.13848
70. Weitz JI, Gross PL. New oral anticoagulants: which one should my patient use? *Hematology Am Soc Hematol Educ Program.* (2012) 2012:536–40. doi: 10.1182/asheducation-2012.1.536
71. Baumann Kreuziger LM, Morton CT, Dries DJ. New anticoagulants: a concise review. *J Trauma Acute Care Surg.* (2012) 73(4):983–92. doi: 10.1097/TA.0b013e318265cf9e
72. Abdelwahed W, Degobert G, Stainmesse S, Fessi H. Freeze-drying of nanoparticles: formulation, process and storage considerations. *Adv Drug Deliv Rev.* (2006) 58(15):1688–713. doi: 10.1016/j.addr.2006.09.017
73. Kim M, Jeong M, Hur S, Cho Y, Park J, Jung H, et al. Engineered ionizable lipid nanoparticles for targeted delivery of RNA therapeutics into different types of cells in the liver. *Sci Adv.* (2021) 7(9):eabf4398. doi: 10.1126/sciadv.abf4398